Project Proposal

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Weather Prediction

Weather Prediction System with MLOps Pipeline flow

Project Description

The goal of this project is to build a machine learning application that predicts whether it will rain based on weather data.

The focus of the project is not building a highly complex ML model, but creating a complete and reproducible ML workflow using the tools learned during the course.

Project Objectives

1. Machine learning model development
2. Data versioning
3. Experiment tracking
4. MLflow usage
5. Docker containerization
6. Continuous Integration (CI)
7. Workflow orchestration
8. Local deployment
9. Monitoring

Problem Statement

The system will predict:

Rain Tomorrow: Yes or No

Using historical weather data such as:

1. Temperature
2. Humidity
3. Wind speed
4. Pressure

Dataset

A small public weather dataset from Kaggle will be used.

Example features:

1. MinTemp
2. MaxTemp
3. Humidity
4. WindSpeed
5. Pressure
6. RainToday

Target:

1. RainTomorrow

Machine Learning Model

An ML model will be implemented using Scikit-learn.

Chosen model:

1. Logistic Regression

Evaluation metrics:

1. Accuracy
2. F1-score

MLOps Components

1. Data Management and Versioning Tool

1. Using **DVC** to version datasets.

2. Experiment Tracking Tool

1. Track experiments using **MLflow**.
2. Store:
   a. Parameters
   b. Metrics

c. Model artifacts

3. Containerization Tool

1. Using **Docker** to containerize the application.

4. Continuous Integration Flow

1. Use **GitHub Actions** to:
    a. Run tests
    b. Validate pipeline
    c. Build a Docker container

5. Orchestration Tool & Flow

Pipelining Tool:

1. Kubeflow Pipeline

Pipeline steps:

1. Data preprocessing
2. Model training
3. Evaluation
4. Saving model

6. Deployment Strategy

1. Deploy model locally using:
    a. FastAPI
    b. Docker

Users can send weather inputs and receive predictions.

7. Monitoring Tool

Monitoring component:

1. Prediction logs
2. Accuracy tracking
3. Input/output logging

System Workflow

1. Dataset
2. DVC Versioning
3. Preprocessing
4. Model Training
5. MLflow Tracking

6. Docker Container
7. GitHub Actions CI
8. FastAPI Deployment
9. Monitoring Logs

Technologies Used

Task	Tool
Programming	Python
ML Library	Scikit-learn
Tracking	MLflow
Versioning	DVC
Containerization	Docker
CI/CD	GitHub Actions
Deployment	FastAPI
Orchestration	Kubeflow / Scripts
Repository	GitHub

Expected Outcome

At the end of the project:

1. A working weather prediction model will be created.
2. The system will be reproducible and deployable.
3. The project will simulate a production ML workflow.